

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	A Roja	Reg. No.:	192424226
Section / Batch:	2024	Date:	01-08-2026

Code of Conduct

I A Roja [192424226] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Roja

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analysing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Answer:

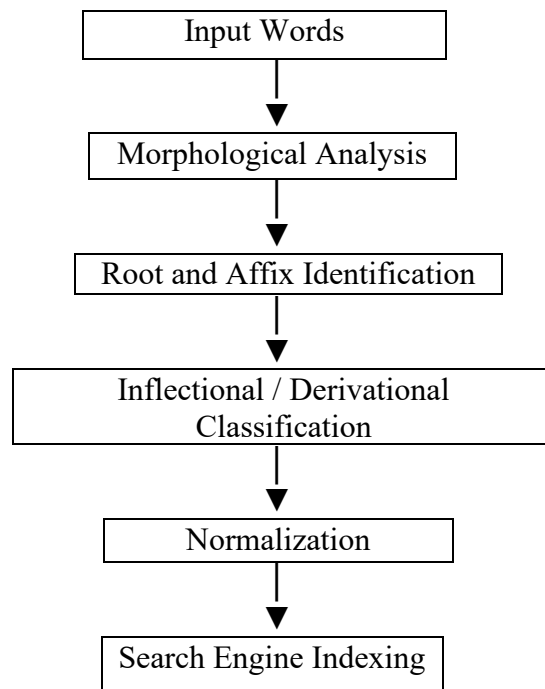
Scenario

A large-scale search engine indexing system processes user queries containing word variants such as "analysing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation.

Morphological Processing Approach

The morphological processing approach analyses each word by separating it into its root and affixes. It then identifies whether the word is inflectional or derivational and normalizes all related words to a common base form. This enables the search engine to retrieve relevant documents even when different word forms are used.

Process



Morphological Breakdown

Input Word	Root	Affix	Type
analysing	analyse	ing	Inflectional
analysis	analyse	sis	Derivational
analytical	analyse	tical	Derivational

Normalized Representation

Input Word Normalized Form

analysing	analyse
analysis	analyse
analytical	analyse

Conclusion

The words are successfully decomposed into their root and affixes. Analysing is an inflectional form, while analysis and analytical are derivational forms. All variants are normalized to analyse, improving search engine indexing and retrieval.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

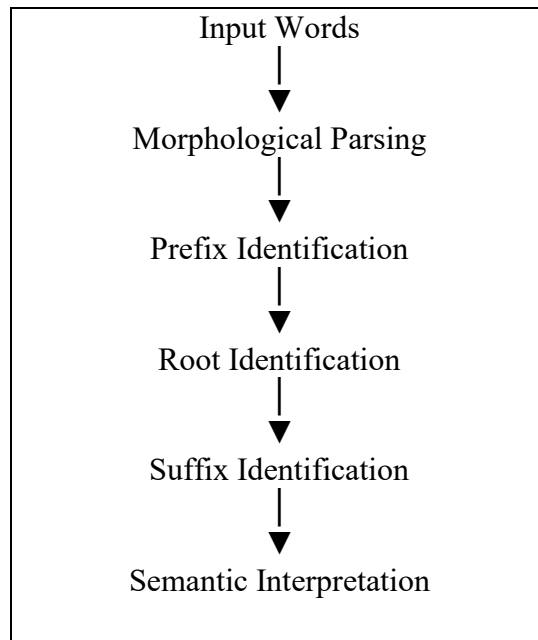
Scenario

An AI-based sentiment analysis platform receives product reviews containing "disagree", "agreement", and "agreeable".

Morphological Parsing Strategy

The parser identifies prefixes, suffixes, and the base form of each word. It analyzes derivational changes to understand how the meaning changes while maintaining semantic consistency during sentiment analysis.

Process



Morphological Breakdown

Input Word	Prefix	Root	Suffix
disagree	dis	agree	—
agreement	—	agree	ment
agreeable	—	agree	able

Classification

Word	Transformation
disagree	Derivational
agreement	Derivational
agreeable	Derivational

Semantic Interpretation

- disagree – expresses the opposite meaning of *agree*.
- agreement – represents the state or act of agreeing.
- agreeable – describes a pleasant or cooperative quality.

Normalized Representation

All words are normalized to:

Agree

Conclusion

The parser correctly identifies prefixes, suffixes, and root words while preserving semantic meaning. All forms are mapped to the common base form agree.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown

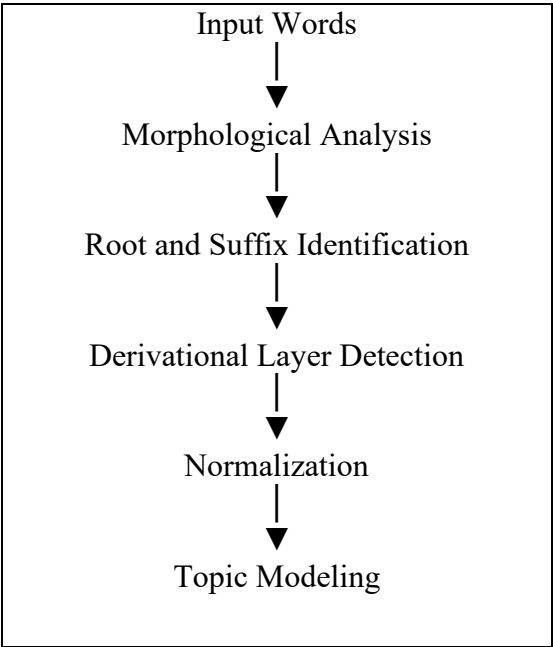
Scenario

A text analytics pipeline processes "govern", "government", and "governance".

Morphology-Based Normalization Mechanism

The normalization mechanism decomposes each word into its root and suffix, identifies derivational layers, and maps all forms to a common base representation for topic modeling and clusterin

Process



Morphological Breakdown

Input Word	Root	Suffix
govern	govern	—
government	govern	ment
governance	govern	ance

Derivational Layers

Word	Type
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govern	Base Form
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government	Derivational
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governance	Derivational
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Normalized Representation

Input Word Normalized Form

govern	govern
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government	govern
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governance	govern
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Conclusion

The morphology-based normalization mechanism maps all word forms to the common root govern, improving document clustering and topic modelling.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyse how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

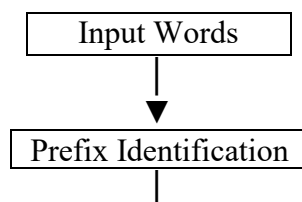
Scenario

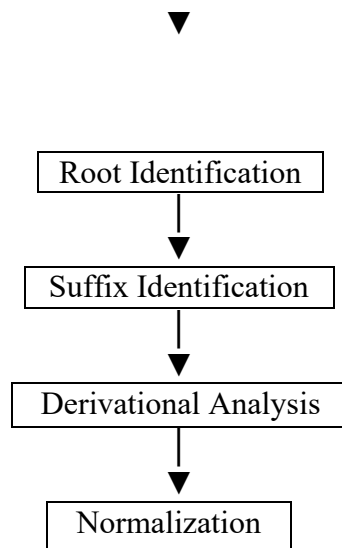
A document classification system processes "activate", "activation", and "reactivation".

Morphological Parsing Strategy

The strategy identifies prefixes, suffixes, and root words while distinguishing derivational layers. It analyzes how affixes modify the meaning and grammatical category of each word

Process





Morphological Breakdown

Input Word Prefix Root Suffix

activate	—	activate	—
activation	—	activate	ion
reactivation	re	activate	ion

Meaning and Word Class

Word Effect

activate	Base verb
activation	Noun formed from the verb
reactivation	Indicates activation again

Normalized Representation

Input Word Normalized Root

activate	activate
activation	activate
reactivation	activate

Conclusion

The parsing strategy successfully identifies prefixes, suffixes, and root forms. All words are normalized to activate, improving semantic grouping and indexing.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word

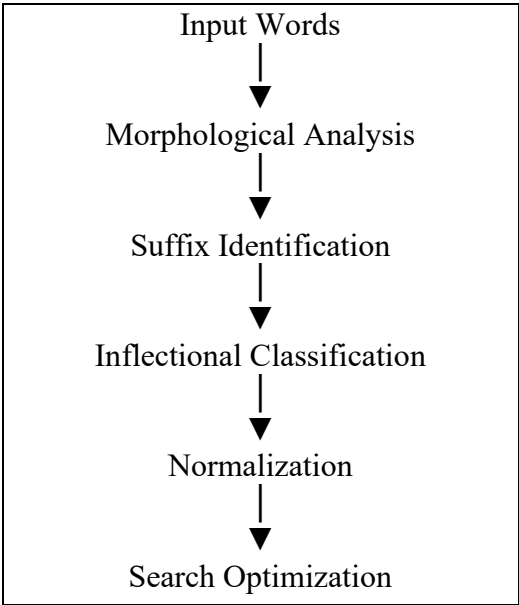
Scenario

A search optimization module processes "create", "creates", and "creating".

Morphology-Based Normalization Approach

The approach focuses on inflectional transformations by identifying suffix patterns and mapping all grammatical variations to the common base form without changing the meaning.

Process



Morphological Breakdown

Input Word	Root	Suffix
create	create	—
creates	create	s
creating	create	ing

Classification

Word Transformation

create Base Form
creates Inflectional
creating Inflectional

Normalized Representation

Input Word Normalized Form

create create

Input Word Normalized Form

creates	create
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creating	create
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Conclusion

The normalization approach successfully identifies suffix patterns, classifies the grammatical transformations as inflectional, and maps all word forms to the common base form create. This improves search optimization by ensuring consistent retrieval across different grammatical variations.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____